FMA 3TC10: Analyse fonctionnelle et de Fourier

Functional analysis and Fourier analysis, in the order of the course poly. The course sets conventions on the extended real line, then works through the topology of normed vector spaces — metric spaces, continuity of linear and bilinear maps, density and separability, completeness and Banach spaces; the Lebesgue integral and the L^p spaces, with their inequalities and convolution; Hilbert spaces, orthogonal projection, Hilbert bases and Fourier series; the Fourier transform on $\mathbb{R}$, first on L^1 and then extended to L^2; regularity, decay at infinity and the Schwartz class; and Fourier analysis on $[0,1[$ and on $\mathbb{Z}$, with the Fejér kernel.

No chapters are written yet: this file is the setup, and chapters are added at the end of it as they are written.

Titles and technical terms are in French, the language of the course and its exam: each term is glossed in English once, where it is defined, definitions keep the course's own French, and the commentary is in English.

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